INCIDENT INVESTIGATION REPORT

AsyncRAT Infection Chain & Technical Forensic Analysis

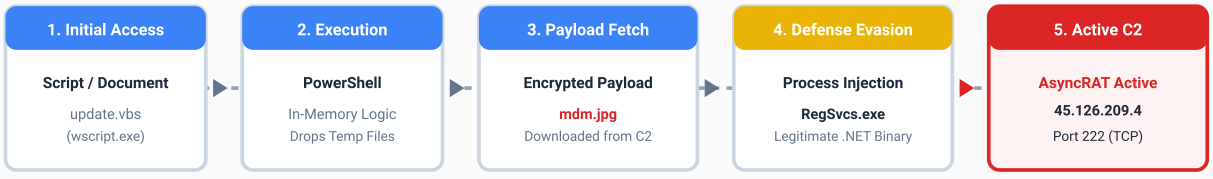
INCIDENT TYPE	MALWARE FAMILY	SEVERITY LEVEL	LEAD ANALYST
Phishing / Script Malware	AsyncRAT	HIGH / CRITICAL	Ahmed Raafat

1. EXECUTIVE SUMMARY

A security investigation was conducted into a multi-stage infection vector delivering **AsyncRAT** remote access malware. The threat actor utilized initial script-based execution via **Windows Script Host (`wscript.exe`)** to bypass standard perimeter security controls. The execution chain leveraged living-off-the-land binaries (LOLBins), specifically invoking **PowerShell** to retrieve secondary malicious payloads (`mdm.jpg`) and ultimately performing process injection into the legitimate .NET utility **`RegSvcs.exe`** . Outbound Command and Control (C2) communication was successfully identified and isolated at target infrastructure **`45.126.209.4:222`** .

2. ATTACK CHAIN TOPOLOGY

The diagram below illustrates the multi-stage execution and defense evasion chain identified during analysis:



1. **Initial Access:** Execution initiated via deceptive file vector, spawning script host binaries.
2. **Script Execution:** Native interpreter `wscript.exe` launches embedded obfuscated PowerShell command strings.
3. **Staging & Dropper:** Secondary PowerShell commands fetch obfuscated binary data disguised as `mdm.jpg` .
4. **Process Injection:** Code is reflectively injected into `RegSvcs.exe` to mask presence behind a trusted system executable.

3. TECHNICAL FORENSIC ANALYSIS

Static and dynamic analysis confirmed that the primary payload is a variant of **AsyncRAT**, an open-source Remote Access Trojan with capabilities including keylogging, credential harvesting, system monitoring, and remote command execution.

Defense Evasion Mechanism: The malware avoids direct file drop onto disk in unencrypted form. By leveraging `RegSvcs.exe` (a Microsoft signed .NET utility), the malicious code executes in memory under the context of a trusted process, bypassing traditional signature-based Endpoint Detection and Antivirus solutions.

4. INDICATORS OF COMPROMISE (IOCS)

Artifact Type	Indicator / Value	Description & Context
File Indicator	`update.vbs`	Initial VBScript launcher executed via WScript.
File Indicator	`mdm.jpg`	Disguised payload containing obfuscated AsyncRAT executable.
Network C2	`45.126.209.4 : 222`	Active Command & Control IP and non-standard TCP port.
Injected Process	`...\Framework\v4.0.30319\RegSvcs.exe`	Target Microsoft .NET binary utilized for process injection.

5. DETECTION & MITIGATION RECOMMENDATIONS

- **LOLBin Process Monitoring:** Implement EDR rules to flag instances where script engines (`wscript.exe`, `cscript.exe`) or PowerShell spawn system binaries like `RegSvcs.exe` or `RegAsm.exe`.
- **Network Perimeter Filtering:** Immediately block outbound connection requests to destination IP `45.126.209.4` across all firewalls and network proxy gateways.
- **PowerShell Logging Enforcement:** Enable PowerShell Script Block Logging (Event ID 4104) and Module Logging to capture de-obfuscated script logic executed in memory.
- **Attack Surface Reduction (ASR):** Block executable content from email attachments and raw script execution for standard non-administrative users.

> **Analyst Conclusion & Containment Status**
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> The threat vector relies heavily on living-off-the-land techniques and social engineering. Immediate isolation of affected host machines, combined with blocking the identified C2 IP address at the network perimeter, effectively neutralizes the threat.